Treatment Success and Failure Prediction Using Machine Learning Models for Pulmonary Tuberculosis Patients Under the DOH-CAR TB-DOTS Program

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Abstract

Pulmonary Tuberculosis (PTB) remains a critical public health challenge in the Cordillera Administrative Region (CAR), Philippines, where complex patient profiles and programmatic gaps contribute to persistent treatment failure. This study develops machine learning models to predict treatment success or failure among PTB patients in DOH-CAR TB-DOTS facilities covering 2015–2025. A standardized preprocessing pipeline performs schema harmonization, pseudonymization, and diagnostic-first missing data handling classifying each variable's missingness mechanism. Supervised ensemble tree-based classifiers—XGBoost, LightGBM, and Random Forest—were trained alongside Recurrent Neural Network and Long Short-Term Memory temporal architectures to capture longitudinal treatment dynamics. Shapley Additive Explanations (SHAP) were applied to quantify the contribution of each variable to model predictions. The validated models are embedded in a Clinical Decision Support System (CDSS) providing dynamic risk estimates and narrative explanations to support early clinical intervention while preserving clinician judgment.

Keywords: tuberculosis, machine learning, treatment outcome prediction, clinical decision support system, explainable AI

Introduction

Pulmonary Tuberculosis (PTB) is a chronic infectious disease caused by Mycobacterium tuberculosis, a slow-growing and acid-fast bacillus that continues to challenge public health. Despite effective therapies, PTB remains widespread due to its strong association with poverty, undernutrition, and comorbidities such as diabetes and HIV, which weaken immune defenses and increase susceptibility. These factors delay diagnosis, complicate treatment, and contribute to disease progression. Globally, TB remains one of the leading causes of death, with 1.5 million deaths and 10 million new cases recorded in 2018, including half a million with rifampicin resistance.

In the Philippines, TB continues to be a major public health concern. The World Health Organization classifies the country as one of the top contributors to the global TB burden, accounting for approximately 6.8 percent of cases. In 2023, around 37,000 Filipinos died from TB out of 739,000 diagnosed cases. In Baguio City, the DOH-CAR reported 437 confirmed TB cases in 2022 along with over 4,405 missing or unreported cases, making the area a priority for surveillance and treatment improvement.

Contributing factors include high population density, internal migration, the city's cool and humid highland climate conducive to transmission, and its role as a tertiary referral center. Patient comorbidities such as diabetes mellitus, HIV infection, and malnutrition impair immune response and reduce treatment completion rates, while poverty-related barriers disrupt adherence to directly observed treatment short-course (DOTS).

Risk Factors for Treatment Failure

TB patients who acquired infection through close contact were shown to be at higher risk of delayed diagnosis and non-adherence. Patients who missed follow-ups had higher risk of reactivation of infection and mortality. HIV co-infection was determined to be a substantial risk factor for treatment failure. A study in the Philippines revealed that 58.82% of 51 patients who experienced treatment failure had at least one comorbidity. Diabetes Mellitus (DM) further increases the risk of treatment failure, relapse, and death compared to non-diabetic patients.

Machine Learning for TB Outcome Prediction

Machine learning offers distinct advantages in modeling complex and non-linear interactions among demographic, clinical, diagnostic, and treatment-related variables. By leveraging algorithms such as logistic regression, Support Vector Machines, and ensemble-based methods including Extreme Gradient Boosting (XGBoost), machine learning can uncover subtle dependencies that contribute to treatment success or failure. Shapley Additive Explanations (SHAP) quantify the contribution of each variable to a model prediction, summarizing both patient-level explanations and overall feature importance.

The limitations of fragmented risk factor studies and capabilities of interpretable machine learning highlight the need for an integrative approach combining multiple dimensions of data. This study adopts a holistic analytical framework integrating descriptive analysis, supervised predictive modeling, and explanation methods to help healthcare providers identify patients at higher risk of treatment failure, understand factors influencing model's risk estimate, and use that information to support early intervention, tailored monitoring, and improved patient management.

Methodology

This study utilizes a retrospective dataset of PTB patients from the CAR covering 2015 to 2025, consolidated by DOH-CAR. After ethical approval, records are pseudonymized and processed using a standardized preprocessing pipeline, including schema harmonization, error correction, and feature scaling, followed by a diagnostic-first missing data handling protocol.

Data Acquisition and Preprocessing

The study draws on two complementary datasets from the CAR: (1) a non-temporal consolidated registry maintained by DOH-CAR comprising 7,389 patient records with cross-sectional baseline features including demographic variables, clinical indicators, and diagnostic results, used to train static tree-based classifiers; and (2) a temporal monitoring dataset of 599 patients with complete month-by-month treatment monitoring records, manually compiled by the study's Pharmacy Co-authors from facility treatment monitoring forms and used to train longitudinal deep learning and temporal tree-based models.

The non-temporal preprocessing pipeline implements schema harmonization to standardize variable formats and units and pseudonymization to enforce patient privacy. Outcome annotation defines the binary target variable: patients classified as 'Cured' or 'Treatment Completed' are annotated as 'Success,' whereas those marked as 'Failed,' 'Died,' or 'Lost to Follow-up' are annotated as 'Failure.' To prevent data leakage, the dataset is partitioned into training and testing sets prior to any data transformation steps.

Missing Data Handling

Clinical TB registries exhibit heterogeneous missingness arising from inconsistent documentation and varying facility capacities. This study implements a diagnostic-first missing data pipeline that classifies each variable's missingness mechanism—Missing Completely at Random (MCAR), Missing at Random (MAR), or Missing Not at Random (MNAR)—and routes the variable to one of four handling strategies: column exclusion, listwise deletion, missing indicator with fill, or stochastic Multivariate Imputation by Chained Equations (MICE). Variables exceeding 80% missingness are excluded entirely. Four clinical vital signs confirmed MAR are imputed using stochastic MICE with an ExtraTreesRegressor as the conditional model.

Supervised Learning Model Development

The study leverages ensemble tree-based algorithms—XGBoost, LightGBM, and Random Forest. Two sampling strategies are compared for each classifier: no resampling with native class-weighting, and SMOTE combined with Edited Nearest Neighbors (SMOTE-ENN), which increases the training-set failure class by 433%. Three progressive feature set versions are evaluated, ranging from 11 core demographic and diagnostic variables to an extended set of 21 features incorporating facility-level geographic and temporal delay variables.

For temporal modeling, a Hybrid Bidirectional LSTM architecture was adopted to model long-term dependencies, with tree-based temporal baselines including XGBoost and LightGBM, all trained on monthly patient monitoring data from Month 0 through Month 7. The dataset is partitioned using stratified 70/20/10 patient-level split into training, test, and validation sets. Model selection employs a composite scoring metric: score = 0.6 × AUC + 0.4 × Recall(Failure), prioritizing discrimination ability while rewarding sensitivity to the minority failure class.

Explainability and Clinical Decision Support

Shapley Additive Explanations (SHAP) are applied to quantify the marginal contribution of each variable to individual model predictions, surfacing the patient-level factors that drive each risk estimate for clinician review. The validated models are embedded within a Clinical Decision Support System (CDSS) that visualizes patient risk levels, highlights contributing factors through SHAP-based explanations, and dynamically updates risk estimates as new follow-up data become available.

Results and Discussion

This section presents the results of machine learning analyses aimed at predicting treatment success and failure among PTB patients under the DOH-CAR TB-DOTS program. Descriptive statistics summarize patient demographics, clinical characteristics, and treatment outcomes, while predictive model performance is evaluated using test and validation datasets. Both static tree-based models under two sampling conditions and temporal models are compared, and feature importance analyses provide interpretable evidence about factors influencing model predictions.

Dataset Characteristics and Class Distribution

The final validated dataset comprised 7,389 complete patient records encompassing demographic, geographic, and clinical features. Overall, the dataset exhibited significant class imbalance, with 86.2% of evaluated cases classified as treatment success and 13.8% as treatment failure. To ensure robust evaluation, the dataset was stratified into 70% training, 20% testing, and 10% validation subsets. Feature engineering expanded the dataset to 15 predictors, integrating spatial identifiers with temporal markers and derived clinical delays. Resampling using SMOTE combined with Edited Nearest Neighbors was applied exclusively to the training set, increasing the minority (failure) class by 433% and achieving a near-balanced 0.70:1 ratio.

Outcome Distribution and Longitudinal Trends

The distribution of treatment outcomes highlighted both programmatic successes and persistent challenges. The vast majority of the cohort achieved positive outcomes, with 65.1% classified as 'Treatment Completed' and 14.2% as 'Cured.' Conversely, adverse outcomes accounted for a critical minority: patients 'Lost to Follow-up' comprised 8.4%, patients who 'Died' accounted for 3.3%, and treatment failure was 0.9%. Analysis of annual treatment outcomes from 2015 to 2025 revealed significant temporal shifts: success rates remained in the 80%–89% range, with 88.92% achieved in 2023. Mortality peaked at 6.63% in 2021, aligning with COVID-19 pandemic disruptions, before declining to 3.66% by 2024. Loss to follow-up rates declined from 11.99% (2020) to 5.56% (2024), suggesting improved patient tracking and adherence interventions.

Figure 1
Distribution of Treatment Outcomes Among PTB Patients

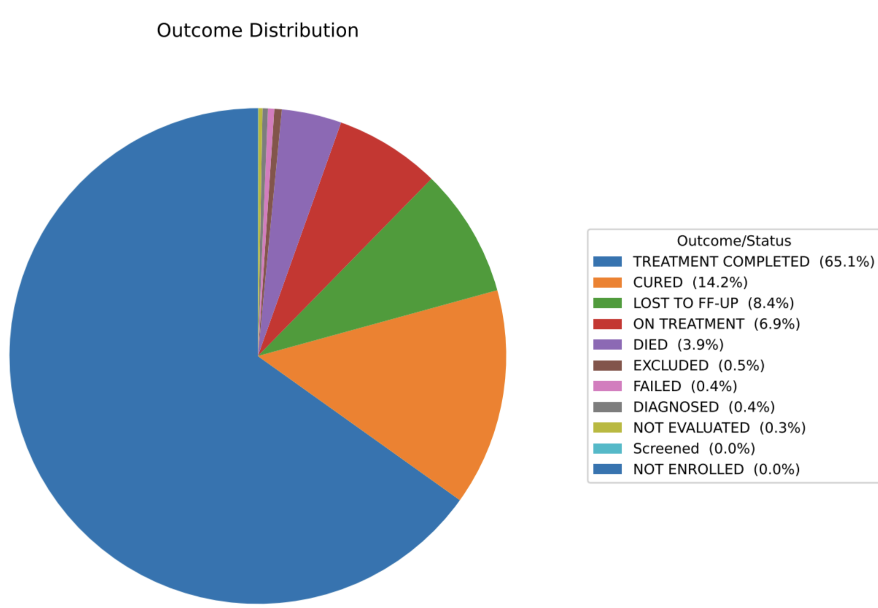


Figure 1 displays the distribution of programmatic treatment outcomes for the entire cohort. The dominance of 'Treatment Completed' and 'Cured' classifications demonstrates the overall effectiveness of the DOH-CAR TB-DOTS program. However, the 8.4% loss to follow-up rate and 3.3% mortality rate underscore the importance of early risk identification and targeted interventions.

An analysis of annual treatment outcomes from 2015 to 2025 revealed significant temporal shifts, reflecting both programmatic improvements and external systemic shocks:

1. Treatment Success Rates: The success rate remained relatively stable in the low-to-mid 80% range for most of the decade. Success rates dipped slightly to 81.03% in 2018 before steadily climbing to a peak of 88.92% in 2023.

2. Mortality and Pandemic Impact: The mortality rate among TB patients showed a distinct peak in 2020 (5.47%) and 2021 (6.63%), up from 3.65% in 2019. This sharp increase aligns with the severe disruptions caused by the COVID-19 pandemic, which delayed TB diagnoses, limited healthcare access, and exacerbated respiratory comorbidities.

3. Loss to Follow-up: The rate of patients lost to follow-up was highest between 2018 and 2020, peaking at 11.99%. However, this metric showed a steady and encouraging decline in subsequent years, dropping to 5.56% by 2024. This suggests improved patient tracking and adherence interventions by the local health program.

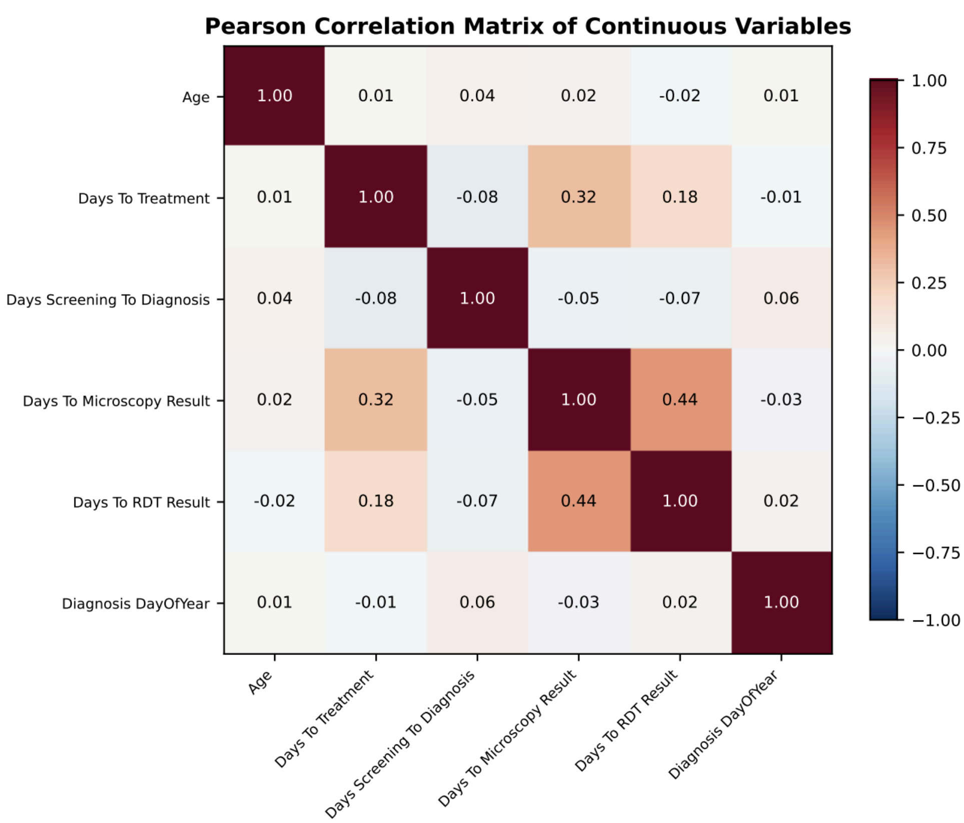
4. 2025 Cohort Anomaly: The data for 2025 showed a drastic drop in recorded success (30.01%) and 52 a massive spike in unresolved cases (64.80%). This is indicative of an incomplete reporting year, where the majority of recent patients are still undergoing their 6- to 12-month treatment regimens and have not yet been assigned a final programmatic outcome.

Diagnostic and Treatment Intervals

The temporal intervals between screening, diagnosis, and treatment initiation were extracted to identify operational delays, which are known risk factors for disease transmission and treatment failure. Median times revealed: (1) screening to diagnosis: 2.0 days (Q3 = 21.09 days); (2) diagnosis to treatment: 2.68 days (Q3 = 13.12 days); and (3) microscopy results: 6.60 days (Q3 = 14.95 days). While median programmatic response is efficient, the long tails represent significant delays for high-risk patients. These interval metrics served as crucial continuous features in the machine learning models, allowing algorithms to correlate logistical delays with eventual probability of mortality or loss to follow-up.

Correlation Analysis

Figure 2
Pearson Correlation Matrix of Continuous Variables



The Pearson correlation analysis revealed weak-to-moderate associations among clinical delay variables. Age exhibited weak negative correlations with time intervals ($|r| < 0.10$), indicating older patients experienced marginally shorter delays. Screening-to-diagnosis and diagnosis-to-treatment intervals showed moderate positive correlation ($r = 0.32$), suggesting that delays at one care cascade stage are associated with delays at subsequent stages. Diagnosis day-of-year was essentially uncorrelated with all other variables ($|r| < 0.05$), confirming seasonal timing is independent of demographic and operational factors.

Static Model Performance

Among all evaluated static models, Random Forest achieved the highest overall performance on the test set.

Table 1
Test Set Model Performance Metrics of Static Classifiers

Model	AUC	Recall (Failure)	F1 (Failure)
Random Forest	0.7154	57.98%	0.3539
Gradient Boosting	0.7121	44.68%	0.3262
LightGBM	0.7119	48.40%	0.3467
XGBoost	0.7095	47.87%	0.3416

Note. AUC = Area Under the Receiver Operating Characteristic Curve.

Random Forest achieved an AUC of 0.7154 and an F1-score of 0.3539 for the failure class, with balanced recall of 57.98% without excessive false positives. Gradient Boosting, LightGBM, and XGBoost achieved comparable AUC values (approximately 0.71), though with slightly lower recall for treatment failure. On the 10% held-out validation set, Random Forest maintained superior performance, achieving a validation AUC of 0.7380 and recall of 52.13% for treatment failure, confirming robustness and suitability for deployment within a clinical decision support framework.

Temporal Model Performance

Table 2 presents the performance metrics of temporal models evaluated on the held-out test set.

Table 2
Test Set Performance Metrics of Temporal Models

Metric	Bi-LSTMBaseline	Bi-LSTMAugmented	XGBoostBaseline	XGBoostSMOTE-ENN	LightGBMSMOTE-ENN	RFSMOTE-ENN
Accuracy	0.9048	0.9048	0.9048	0.9048	0.9048	0.7143
Precision	0.9048	0.9048	0.9048	0.9048	0.9048	0.8824
Recall	1.0000	1.0000	1.0000	1.0000	1.0000	0.7895
F1 Score	0.9500	0.9500	0.9500	0.9500	0.9500	0.8333
ROC-AUC	0.4737	0.3684	0.5789	0.1053	0.2105	0.5263
PR-AUC	0.8847	0.9099	0.9501	0.8356	0.8500	0.9419

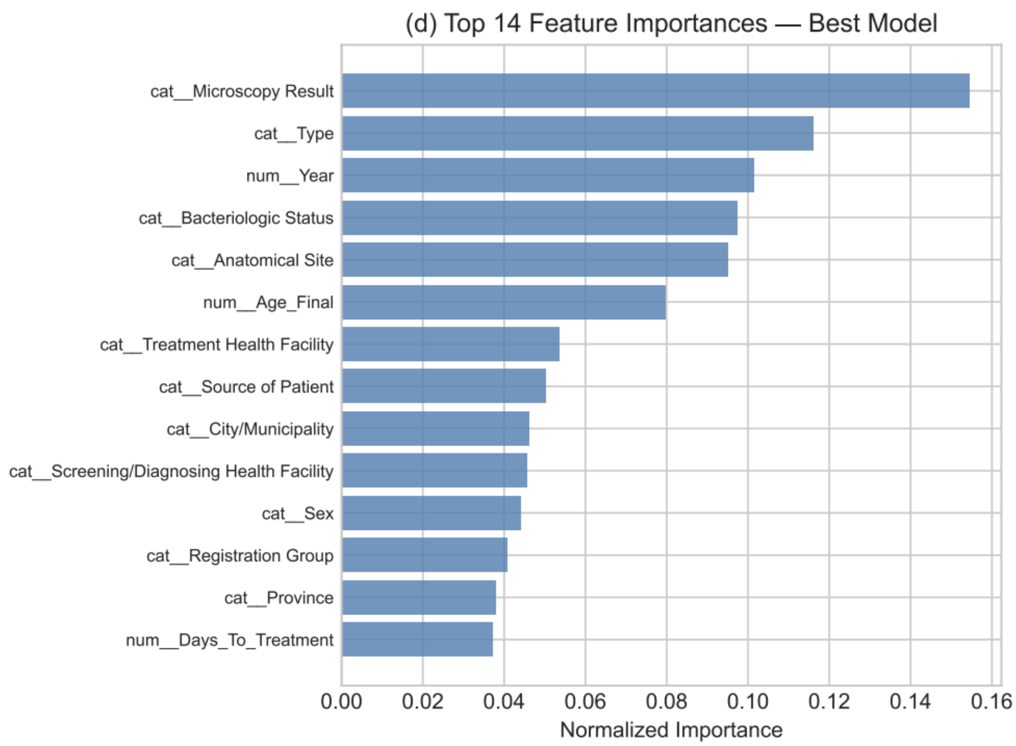
Note. Bi-LSTM = Bidirectional Long Short-Term Memory; RF = Random Forest; ROC-AUC = Area Under Receiver Operating Characteristic Curve; PR-AUC = Area Under Precision-Recall Curve.

Both Bi-LSTM variants and XGBoost Baseline achieved the highest overall accuracy (0.9048) and F1 score (0.9500), reflecting their ability to correctly predict the dominant Success class. However, specificity remained 0.0, demonstrating that no actual Failure cases were correctly identified in the small test set. Random Forest with SMOTE-ENN showed more balanced performance, achieving slightly lower overall accuracy (0.7143) but higher recall for the Failure class (0.7895) and competitive PR-AUC (0.9419), indicating better identification of high-risk patients. These results suggest that while Bi-LSTM and XGBoost excel at predicting the majority outcome, Random Forest with SMOTE-ENN offers the most effective detection of treatment failure, making it more suitable for clinical decision support applications.

Feature Importance Analysis and SHAP Explanations

To understand which patient-level factors most strongly influence treatment outcomes, we analyzed feature importance across the ensemble models using both model-native importance metrics and post-hoc SHAP values. Figure 3 presents the top 15 features ranked by mean absolute SHAP values for the Random Forest model, which integrates local explanations from all patients in the validation set.

Figure 3
Top 15 Features Ranked by Mean Absolute SHAP Values (Random Forest Model)



The SHAP analysis reveals that diagnostic and treatment intervals—particularly days from diagnosis to treatment—emerge as the strongest global predictor of treatment outcomes. Monthly adherence metrics (doses taken, percentage adherence) in mid-course months (M2–M4) are also highly influential, indicating that early adherence patterns are predictive of final treatment success or failure. Clinical and demographic factors such as bacteriologic status, anatomical site, age, and comorbidities contribute meaningfully but with lower average impact than operational delays and adherence metrics.

Patient-Level SHAP Explanations and Risk Stratification

Beyond global feature importance, SHAP provides individual patient-level explanations that clinicians can review to understand why the model assigned a particular risk score. Figure 5 illustrates a hypothetical example of how SHAP values are visualized for a single high-risk patient, showing which factors pushed the model's prediction toward treatment failure and by how much.

Clinical Decision Support System Implementation

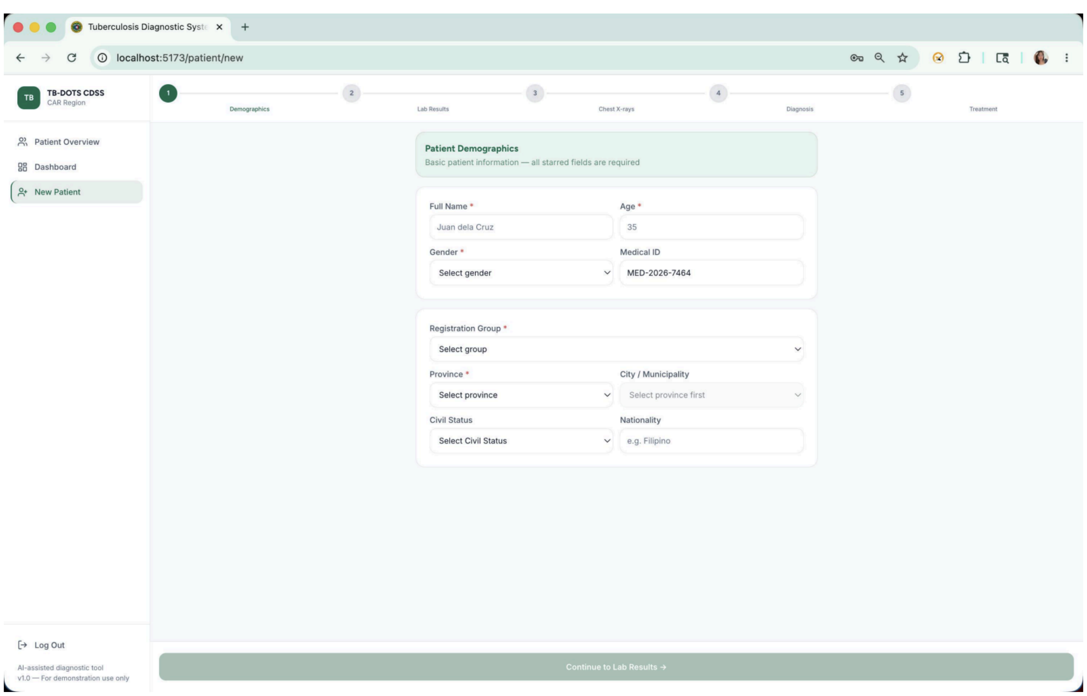
The validated machine learning models are embedded into a Clinical Decision Support System (CDSS) designed for healthcare providers in TB-DOTS facilities. The system is structured around a clear separation of responsibilities: the predictive model estimates the probability of treatment success or failure, the SHAP layer identifies the factors contributing to the model's prediction, the interface visualizes risk and contributing factors, and the clinician interprets these outputs in the context of patient care.

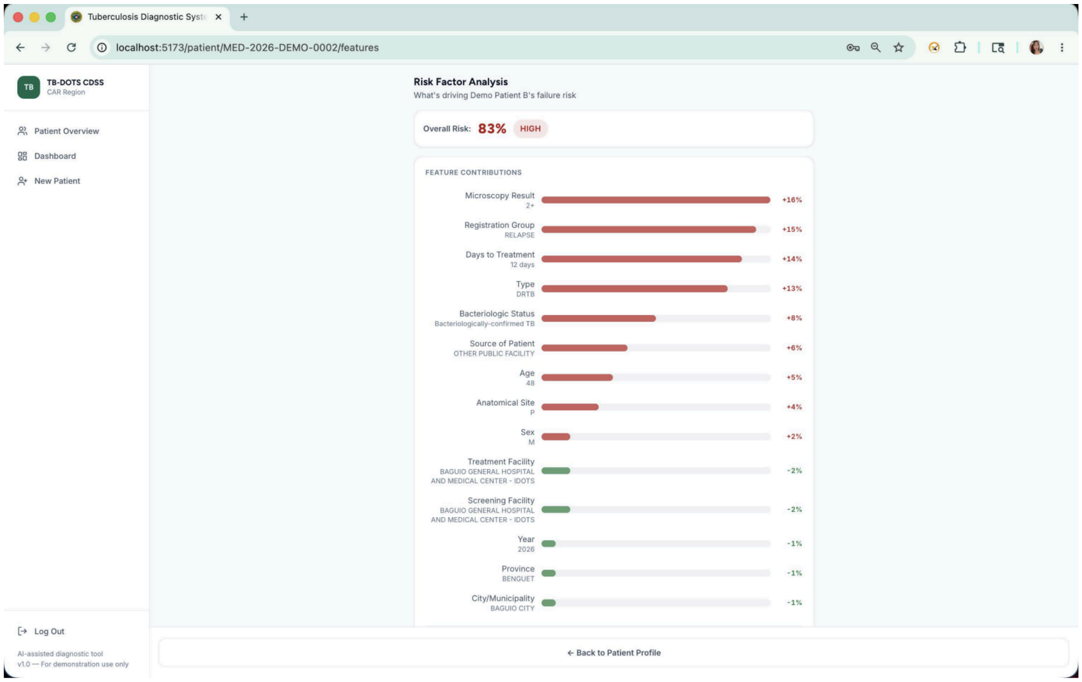
Key Features of the CDSS

The Clinical Decision Support System (CDSS) is designed as a risk stratification and explanation tool to assist healthcare providers rather than act as an autonomous diagnostic or treatment-prescribing system, ensuring that final clinical judgment remains with the treating professional. At its core, the system features a predictive dashboard that displays the probability of treatment failure and success using color-coded risk stratification, where green signifies low risk <20%, yellow indicates medium risk 20% - 50%, and red denotes high risk >50%, allowing providers to easily prioritize high-risk patients for closer monitoring and targeted interventions. To provide transparency, a risk factor visualization layer utilizes SHAP-based feature contribution visualizations to highlight specific variables driving the risk score—such as adherence gaps, laboratory results, or diagnostic delays—enabling clinicians to inspect exactly which patient factors the model identified as concerning. This quantitative data is further contextualized by a narrative explanation layer, where a constrained small language model verbalizes the SHAP-based attributions into readable clinical text, explaining, for instance, how a seven-day delay in treatment initiation or low adherence in a specific month quantitatively elevates failure risk. Finally, the CDSS supports dynamic updates by accepting new follow-up data—including monthly weight, smear results, and adherence percentages—and recomputing risk scores in real time to reflect the patient's current clinical status throughout the entire duration of their treatment.

The CDSS is framed as a risk stratification and explanation tool, not as an autonomous diagnostic or treatment-prescribing system. Final clinical judgment remains with the treating healthcare professional.

Figure 4
Clinical Decision Support System Workflow and User Interface Mockup





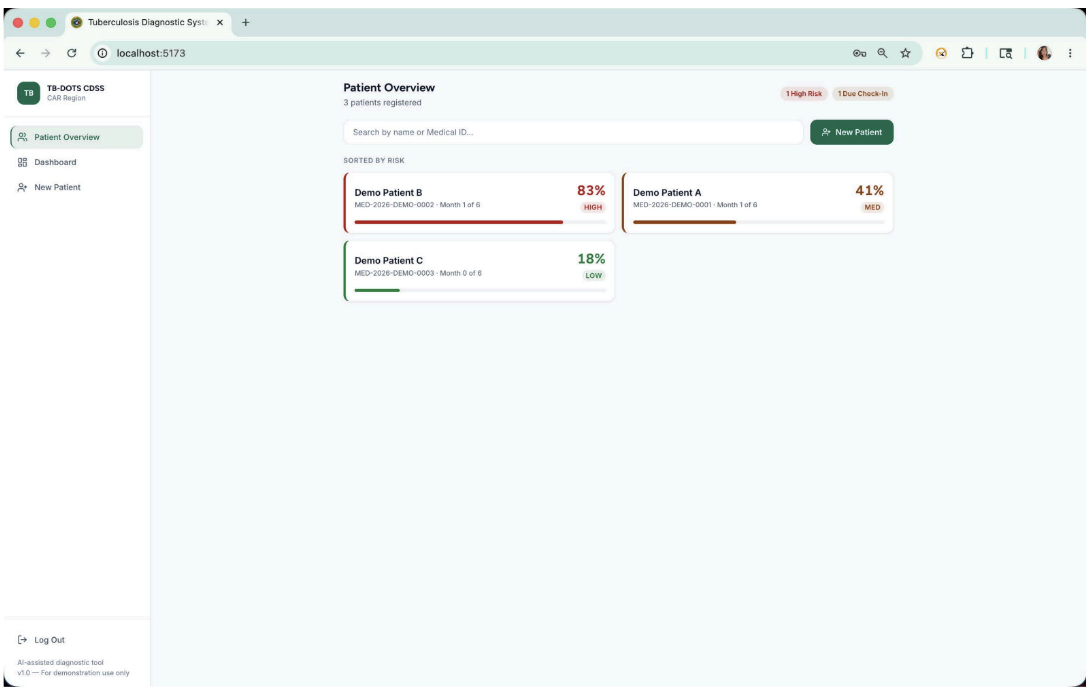


Figure 4 illustrates the integrated workflow of the CDSS. Patient data flows into the predictive model, which generates a treatment failure risk probability. SHAP explanations are automatically computed and visualized, showing the contributing factors. The healthcare provider reviews both the risk score and the contributing factors, and then makes clinical decisions informed by but not dictated by the system.

Model Generalization and Clinical Utility

The validation set performance (AUC = 0.7380, Recall = 52.13% for failure class) demonstrates that Random Forest with SMOTE-ENN generalizes well beyond the training cohort. This level of discriminative performance, combined with interpretable SHAP explanations, provides clinically useful support for identifying high-risk patients. The 52% recall for the failure class indicates that the model captures approximately half of all patients who will experience treatment failure, enabling earlier identification and intervention compared to standard paper-based monitoring.

The moderate AUC (0.7380) reflects the inherent difficulty of predicting treatment failure from baseline clinical and operational data: many patients with similar risk profiles have disparate outcomes due to unmeasured social determinants, medication side effects, or unpredictable treatment adherence. However, the model's ability to identify a substantial minority of high-risk patients, combined with interpretable explanations of why they are flagged, provides meaningful value for TB control programs aiming to concentrate limited resources on the most vulnerable populations.

Conclusions

This study investigated the use of machine learning models to predict treatment outcomes among PTB patients under the DOH-CAR TB-DOTS program, using monthly patient monitoring data alongside baseline clinical features. The test set performance revealed that while Bi-LSTM and XGBoost Baseline models achieved high overall accuracy (0.9048) and F1 scores (0.9500), their ability to detect treatment failures was limited by zero specificity for the minority Failure class. In contrast, Random Forest with SMOTE-ENN demonstrated more balanced performance, achieving a recall of 0.7895 for treatment failure and a high PR-AUC (0.9419), indicating greater reliability in identifying patients at risk of adverse outcomes.

The study demonstrates that integrating temporal monitoring data with static baseline features improves predictive performance over static data alone. The application of SMOTE-ENN effectively addresses the extreme class imbalance inherent in TB treatment outcomes, enhancing the model's capacity to detect high-risk patients. Shapley Additive Explanations prove invaluable in bridging the gap between predictive performance and clinician interpretability: by surfacing the specific patient-level factors driving each risk prediction, SHAP enables healthcare providers to understand and trust the model's outputs.

Predictive models—particularly Random Forest with SMOTE-ENN—have practical value for clinical decision support systems, enabling clinicians to identify patients at risk of treatment failure early and prioritize interventions accordingly. The embedded CDSS, with its real-time risk updates and narrative explanations, provides a scalable tool for TB program managers to strengthen treatment outcomes. Overall, this study aligns with Sustainable Development Goal 3 – Good Health and Well-Being by providing a data-driven approach to improving TB treatment outcomes in the Cordillera Administrative Region.

Recommendations

Based on the findings, it is recommended that healthcare facilities and policymakers consider integrating machine learning models like Random Forest with SMOTE-ENN into existing TB monitoring programs to improve treatment outcomes and optimize resource allocation. Future research should focus on larger, multi-region datasets to enhance model generalization and explore hybrid modeling approaches combining deep learning and tree-based methods. Investigations into additional features such as social determinants of health and geographic risk factors are encouraged. Enhancing model interpretability—particularly of Bi-LSTM attention outputs—could improve clinician trust and adoption. Standardizing digital record-keeping practices across TB-DOTS facilities will support development of more comprehensive longitudinal datasets for future studies and enable real-time implementation of the CDSS at scale.

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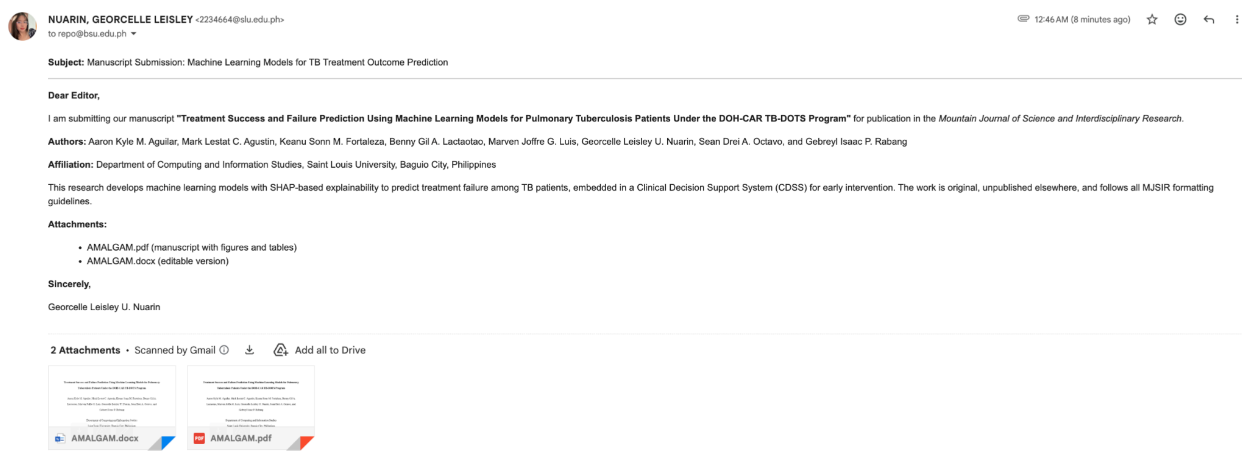
collaboration and contributions of our partner researchers from the SONAHBS Department of Pharmacy, whose expertise significantly enriched this study.

Appendix A: *Proof of Submission to a Journal Publication*


NUARIN, GEORCELLE LEISLEY <2234664@slu.edu.ph>
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Subject: Manuscript Submission: Machine Learning Models for TB Treatment Outcome Prediction

Dear Editor,

I am submitting our manuscript **"Treatment Success and Failure Prediction Using Machine Learning Models for Pulmonary Tuberculosis Patients Under the DOH-CAR TB-DOTS Program"** for publication in the *Mountain Journal of Science and Interdisciplinary Research*.

Authors: Aaron Kyle M. Aguilar, Mark Lestat C. Agustin, Keanu Sonn M. Fortaleza, Benny Gil A. Lactaotao, Marven Joffre G. Luis, Georcelle Leisley U. Nuarin, Sean Drei A. Octavo, and Gebreyl Isaac P. Rabang

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This research develops machine learning models with SHAP-based explainability to predict treatment failure among TB patients, embedded in a Clinical Decision Support System (CDSS) for early intervention. The work is original, unpublished elsewhere, and follows all MJSIR formatting guidelines.

Attachments:

- AMALGAM.pdf (manuscript with figures and tables)
- AMALGAM.docx (editable version)

Sincerely,

Georcelle Leisley U. Nuarin

2 Attachments • Scanned by Gmail · Add all to Drive

AMALGAM.docx · AMALGAM.pdf

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